

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-M6 :Non-right angled Trigonometry

Section I –Multiple Choice Questions (1 Mark each)

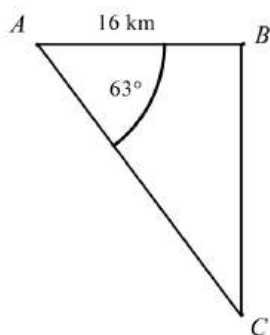
Section II – Longer Questions (Marks indicated beside questions)

Section I**15 marks**

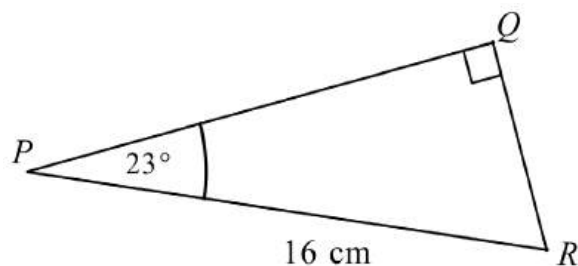
1. A railway track climbs 20m over a horizontal distance of 250m. The angle of elevation is closest to:

- (A) 2°
(B) 5°
(C) 88°
(D) 85°

2. Find the size of the side AC , to the nearest km, in the triangle ABC below.

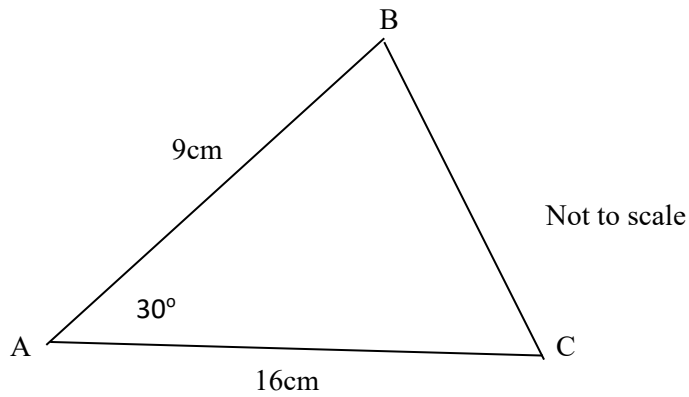


- (A) 7 km
(B) 31 km
(C) 35 km
(D) 228 km
3. Calculate the distance QR in the triangle shown, correct to the nearest millimetre.



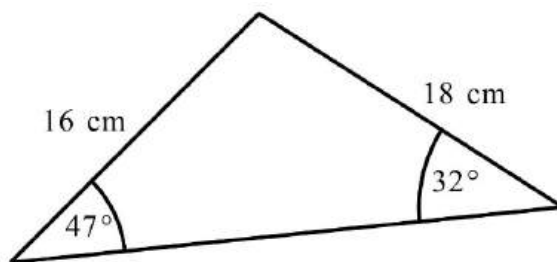
- (A) 6.3 cm
(B) 14.7 cm
(C) 37.7 cm
(D) 40.9 cm

4.



The area of ΔABC is:

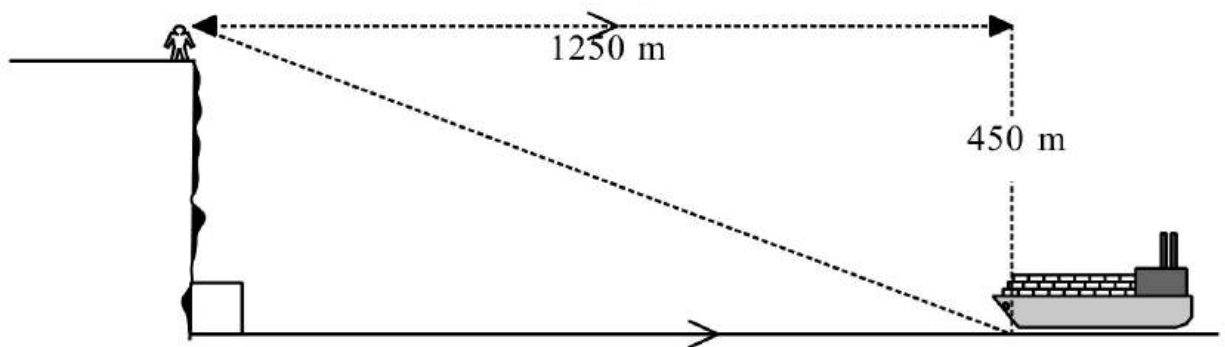
- (A) 18cm^2
 - (B) 36cm^2
 - (C) 72cm^2
 - (D) 144cm^2
5. What is the area of the triangle shown? (Correct to 2 significant figures.)



- (A) 140 cm^2
- (B) 76 cm^2
- (C) 280 cm^2
- (D) 110 cm^2

6. What is the angle of depression of the ship from the viewer on the cliff?

(Answer to the nearest minute)



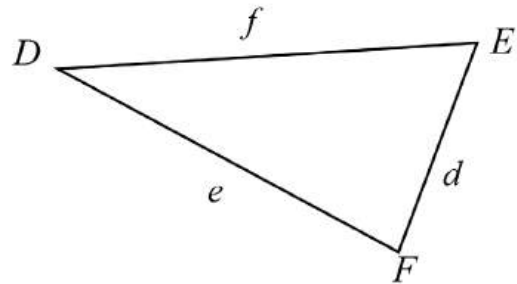
- (A) $19^{\circ} 8'$
 (B) $19^{\circ} 48'$
 (C) $20^{\circ} 8'$
 (D) $21^{\circ} 48'$
7. Which statement is **not** correct for the triangle shown?

(A) $\frac{e}{\sin E} = \frac{f}{\sin F}$

(B) $d = \frac{e \sin D}{\sin F}$

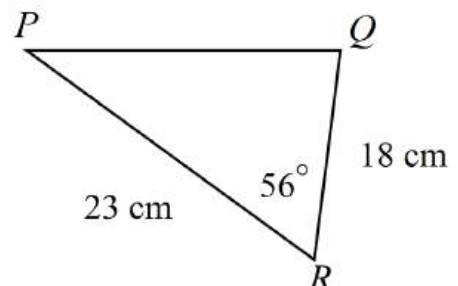
(C) $\frac{\sin D}{d} = \frac{\sin E}{e}$

(D) $e = \frac{f \sin E}{\sin F}$



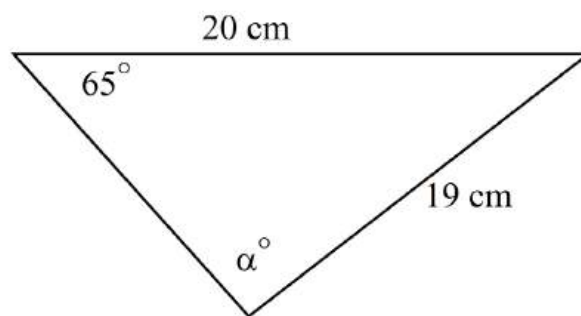
8. Use the cosine rule to find the length of side PQ .

- (A) 12.9 cm
 (B) 14.0 cm
 (C) 19.7 cm
 (D) 44.2 cm



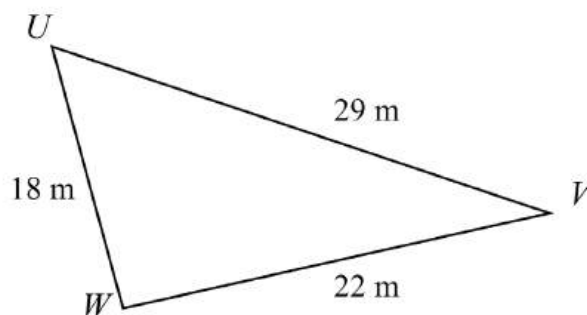
9. Use the sine rule to find the value of α .

- (A) 40
(B) 44
(C) 51
(D) 73



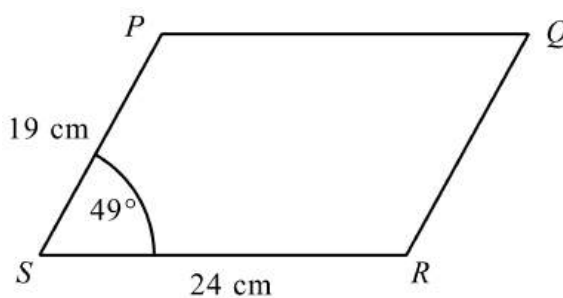
10. Use the cosine rule to find the size of $\angle UVW$.

- (A) 38°
(B) 41°
(C) 49°
(D) 92°

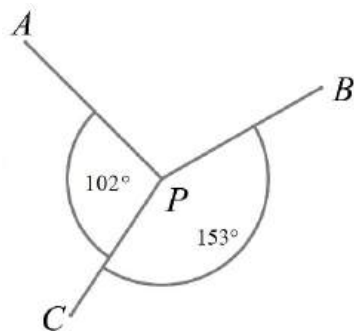


11. Use the triangle area formula to find the area of parallelogram $PQRS$.

- (A) 300 cm^2
(B) 344 cm^2
(C) 456 cm^2
(D) 688 cm^2



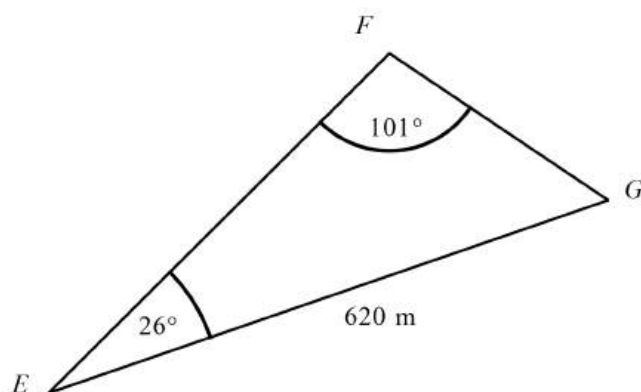
12. In the diagram, A is northwest of P , $\angle APC = 102^\circ$ and $\angle CPB = 153^\circ$.



What is the bearing of B from P ?

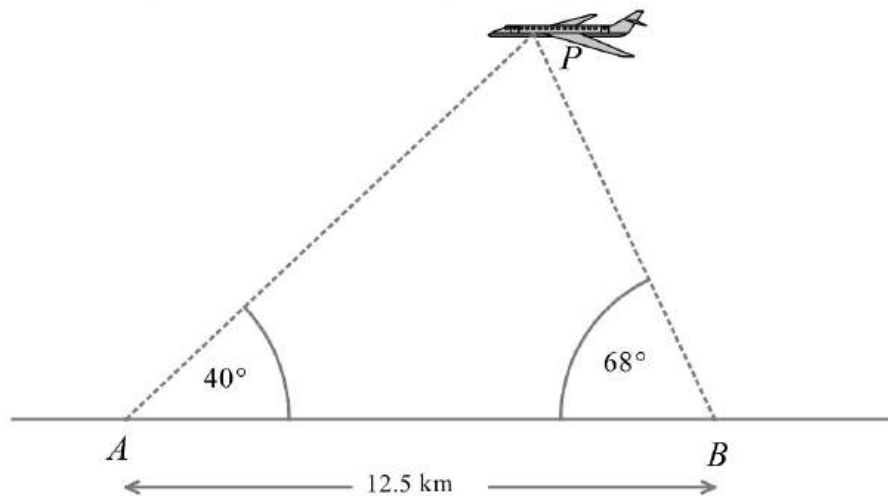
- (A) 045°
 - (B) 060°
 - (C) 213°
 - (D) 315°
13. Find the length of FG .

- (A) 109 m
- (B) 272 m
- (C) 277 m
- (D) 608 m



14. From two points A and B on level ground, the angles of elevation of a plane P are measured to be 40° and 68° respectively.

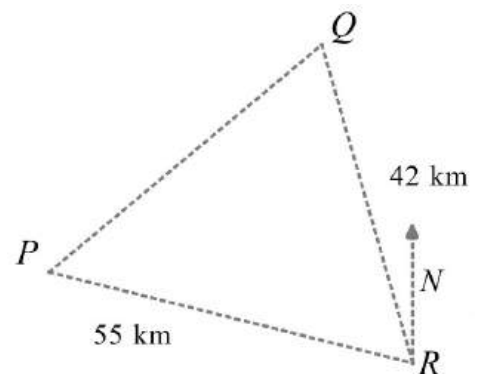
Calculate the distance from A to the plane (AP).



- (A) 7.4 km
(B) 8.1 km
(C) 11.6 km
(D) 12.2 km
15. Three towns P , Q and R lie on a flat plain.
From R , the bearing of Q is 350° and of P is 282° .
 $PR = 55 \text{ km}$ and $QR = 42 \text{ km}$.

Calculate the distance PQ correct to 3 s.f..

- (A) 32.3 km
(B) 50.4 km
(C) 55.2 km
(D) 69.2 km



Western Mathematics

Mathematics Standard 2

MS-M6 :Non-right angled Trigonometry

TOPIC REVISION QUESTIONS - Section II

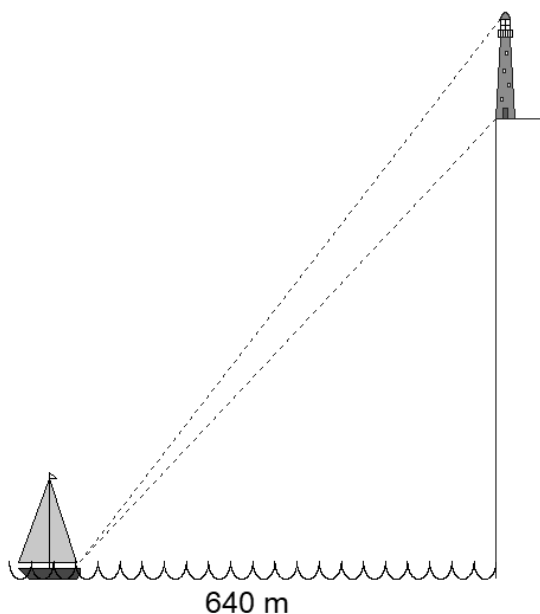
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 16 (4 marks)**Marks**

A ship is 640 metres from the base of a vertical cliff, which has a lighthouse at the top.

The angle of elevation of the top of the cliff is 60° and the angle of elevation of the top of the lighthouse is 65° .



- (a) Calculate the height of the cliff.

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- (b) Calculate the height of the lighthouse.

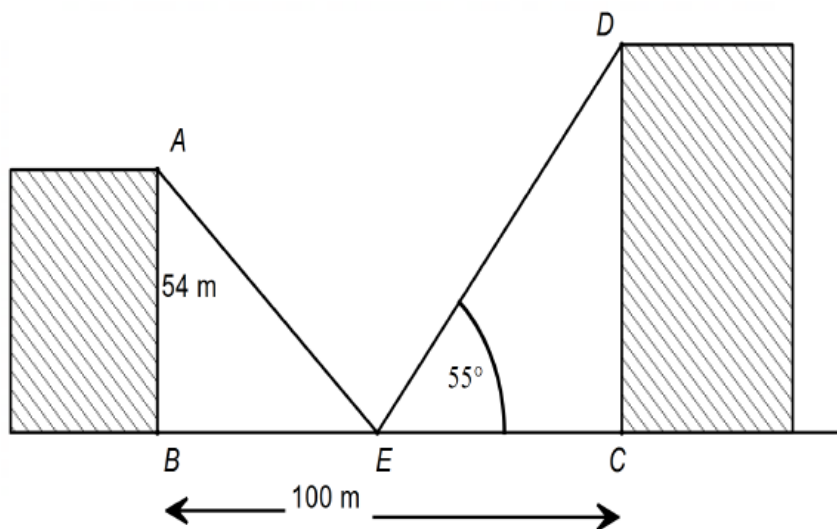
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Question 17 (3 marks)

Evelyn stands at E, halfway between two buildings which are 100 m apart. AB (the height of the building on the left) has been measured to be 54 m. From E, the angle of elevation of D (the top of the building on the right) is 55° .



- (a) What is the angle of elevation of A from E to the nearest minute?

1

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- (b) Evelyn is helping to attach a wire from E to D. Find the length of this wire (DE). (correct to 1 decimal place).

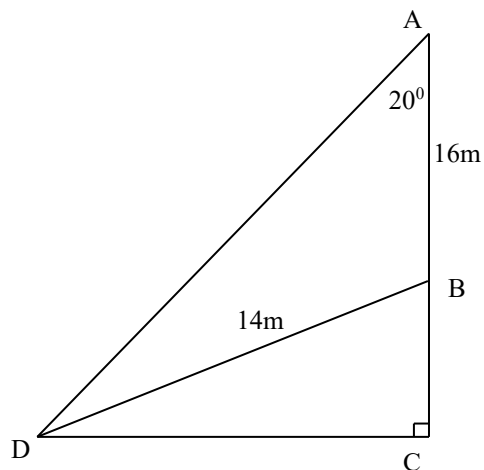
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Question 18 (5 marks)

In $\triangle ACD$, $\angle DAC = 20^\circ$.

B is a point on AC so that $AB = 16$ cm and $DB = 14$ cm.



(a) Find the size of $\angle ADB$.

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(b) Find the size of $\angle DBC$.

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(c) Find the distance AC .

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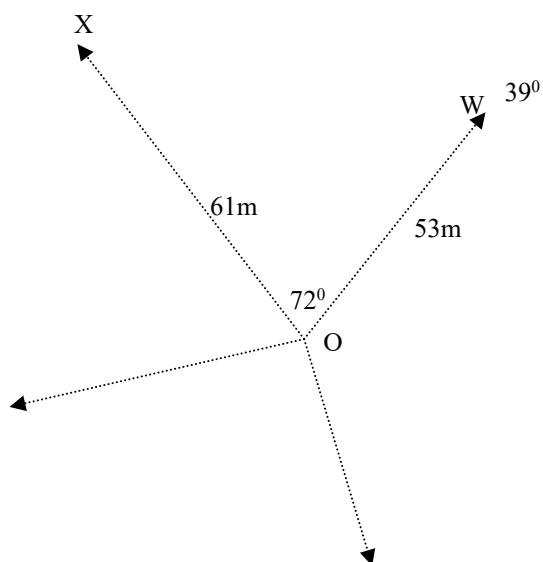
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Question 19 (3 marks)

The diagram shows some of the results of a radial survey of a field.

The bearing of W from O is 039° .

$OW = 53\text{m}$ and $OX = 61\text{m}$ $\angle XOW = 72^\circ$



- (a) Calculate the bearing of X from O . 1

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- (b) Calculate the distance XW . Answer to the nearest metre. 2

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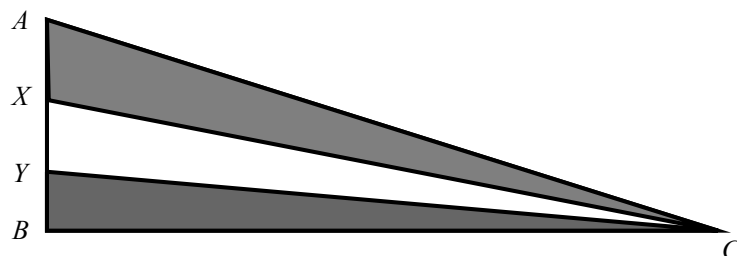
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Question 20 (6 marks)

The triangular banner shown is in three sections.

The triangle ABC is right angled with $BC = 120$ cm and $AB = 45$ cm.

The distances AX , XY and YB are equal and $\angle XYC = 95^\circ$.



- (a) Find the distance YC correct to 3 significant figures.

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- (b) Hence find the area of the triangle XYC .

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- (c) What percentage of the area of the banner is the triangle XYC ?

2

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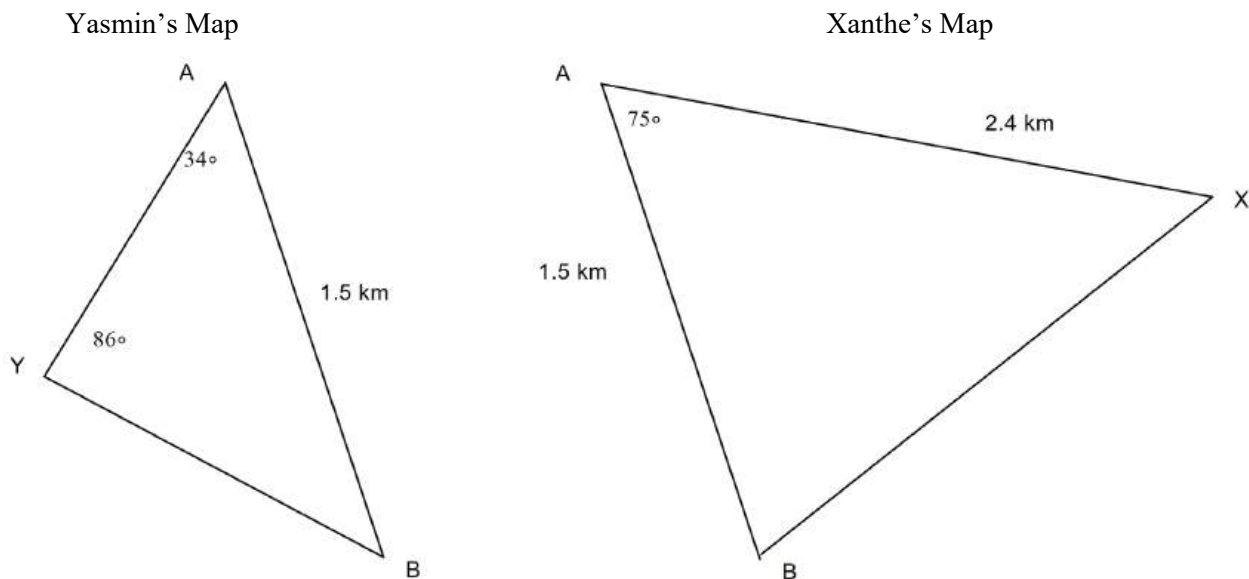
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Question 21 (5 marks)

Xanthe and Yasmin are on a bushwalk. At one stage they become separated. They contact one another by mobile phone and realise they can both see the tops of two towers A and B which are 1.5 km apart. They take bearings with their compasses and mark the information below on their maps. They decide to meet up at tower B.



- (a) How far must Yasmin walk to get to the base of Tower B? 2

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- (b) How far must Xanthe walk to get to the base of Tower B? 2

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- (c) If Tower B is on a bearing 160° from Tower A, what bearing must Yasmin walk along to reach Tower B? 1

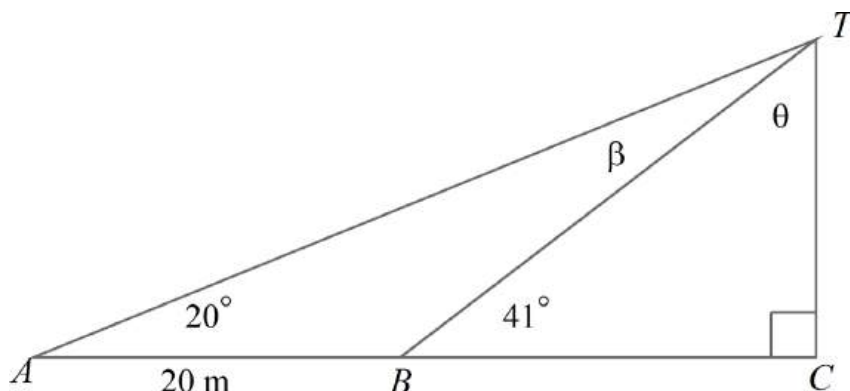
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Question 22 (5 marks)

Hannah wants to calculate the height of one of the tallest trees on her property. From a point A, on level ground, she measures the angle of elevation of the top of the tree (T) as 20° . She then walks 20 metres directly toward the tree to a point B and measures the angle of elevation as 41° .



- (a) Calculate the size of the angles marked θ and β .

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- (b) Use the sine rule to calculate the distance BT, correct to the nearest metre.

2

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- (c) Hence calculate the height of the tree, TC.

2

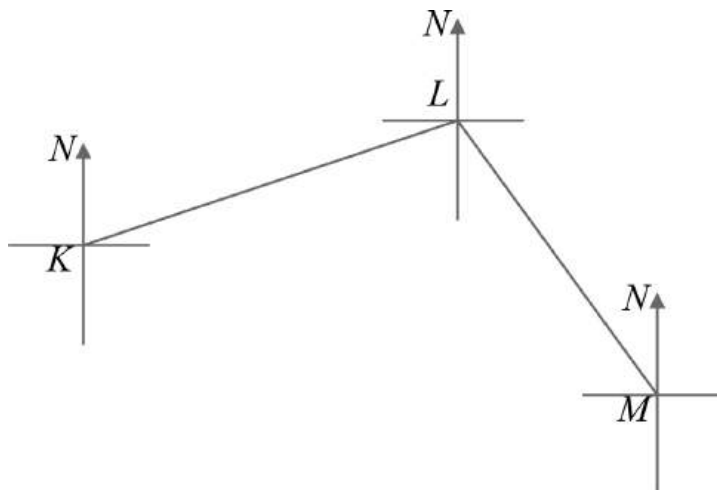
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Question 23 (6 marks)

A plane leaves Kinton (K) and flies on a course 075° for a distance of 120 km to Lowe(L). It then flies on course of 160° for a distance of 140 km to Macy (M).

- (a) Mark the information from above on the sketch. 1



- (b) Calculate the size of $\angle KLM$. 1

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- (c) Calculate the distance KM . 2

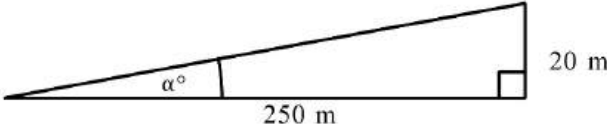
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- (d) Find the bearing of M from K . 2

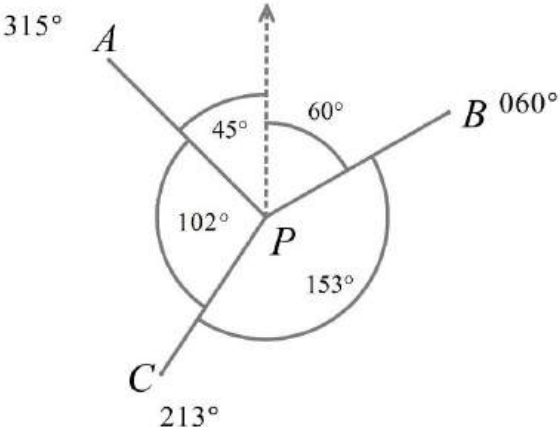
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End of Revision

Western Mathematics
MS-M6 :Non-right angled Trigonometry Revision
Section I - Worked Solutions

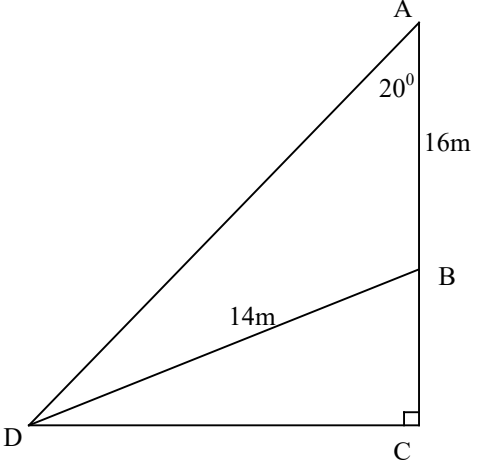
No	Working	Answer
1	 $\tan \alpha = \frac{20}{250}$ $\alpha = \tan^{-1}(0.08)$ $= 4.573921$ $= 5^\circ \text{ (nearest degree)}$	B
2	$\cos 63^\circ = \frac{16}{AC}$ $AC = \frac{16}{\cos 63^\circ}$ $= 35.24302824$ $= 35 \text{ km}$	C
3	$\sin 23^\circ = \frac{QR}{16}$ $QR = 16 \times \sin 23^\circ$ $= 6.25169806$ $= 6.3 \text{ cm}$	A
4	$\text{Area} = \frac{1}{2} b c \sin A$ $= \frac{1}{2} \times 9 \times 16 \times \sin 30^\circ$ $= 36 \text{ cm}^2$	B
5	$\text{Missing angle} = 180 - (47 + 32) = 101^\circ$ $\text{Area} = \frac{1}{2} a b \sin C$ $= \frac{1}{2} \times 16 \times 18 \times \sin 101^\circ$ $= 141.35431441 \dots$ $= 140 \text{ cm (2 sig fig)}$	A
6	$\tan \beta = \frac{450}{1250}$ $\beta = \tan^{-1}\left(\frac{450}{1250}\right)$ $= 19.798876354524929311671766058437$ $= 19^\circ 48'$	B

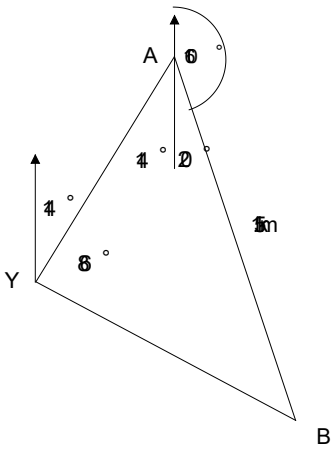
7	Using the sine rule for the triangle $\frac{d}{\sin D} = \frac{e}{\sin E} = \frac{f}{\sin F}$ Taking last 2 sections $\frac{e}{\sin E} = \frac{f}{\sin F}$ so A is correct $d = \frac{e \sin D}{\sin F}$ would come from $\frac{d}{\sin D} = \frac{e}{\sin F}$ so B is incorrect Taking reciprocal of first 2 sections $\frac{\sin D}{d} = \frac{\sin E}{e}$ so C is right Taking $\frac{e}{\sin E} = \frac{f}{\sin F}$ and $\times \sin E$ gives $e = \frac{f \sin E}{\sin F}$ so D is good	B
8	$a^2 = b^2 + c^2 - 2bc \cos A$ $a^2 = 18^2 + 23^2 - 2 \times 18 \times 23 \times \cos 56^\circ$ $= 389.98827596$ $a = \sqrt{389.98827596} = 19.74812082$ $= 19.7 \text{ cm (3 s f)}$	C
9	$\frac{\sin A}{a} = \frac{\sin B}{b}$ $\frac{\sin \alpha}{20} = \frac{\sin 65^\circ}{19}$ $\sin \alpha = \frac{20 \times \sin 65^\circ}{19} = 0.9540082$ $\alpha = \sin^{-1}(0.9540082)$ $= 72.555579$ $= 73^\circ \text{ (nearest degree)}$	D
10	$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ $\cos A = \frac{22^2 + 29^2 - 18^2}{2 \times 22 \times 29}$ $= 0.78448276$ $A = \cos^{-1}(0.78448276)$ $= 38.32713$ $= 38^\circ$	A
11	$\text{Area Triangle} = \frac{1}{2} a b \sin C$ $= \frac{1}{2} \times 19 \times 24 \times \sin 49^\circ = 172.0737843$ $\text{Area Parallelogram} = 172.0737843 \times 2$ $= 344.1475686$ $= 344 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$	B

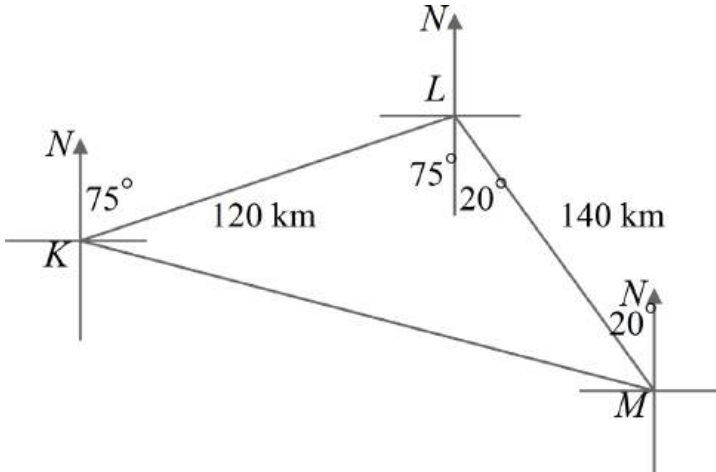
12	Bearing of A from P is 315° Bearing of B = $315^\circ - (102^\circ + 153^\circ)$ $= 315^\circ - 255^\circ$ $= 060^\circ$ 	
13	$\frac{e}{\sin E} = \frac{f}{\sin F}$ $\frac{e}{\sin 26^\circ} = \frac{620}{\sin 101^\circ}$ $e = \frac{620 \times \sin 101^\circ}{\sin 26^\circ}$ $= 276.87712363$ $= 277 \text{ m}$	C
14	$\angle APB = 180 - (68 + 40) = 72^\circ$ $\frac{\overline{AP}}{\sin 68^\circ} = \frac{12.5}{\sin 72^\circ}$ $\overline{AP} = \frac{12.5 \sin 68^\circ}{\sin 72^\circ}$ $= 12.18623498$ $= 12.2 \text{ km (nearest 100 m)}$	D
15	$\angle PRQ = 350 - 282 = 68^\circ$ $PQ^2 = 55^2 + 42^2 - 2 \times 55 \times 42 \times \cos 68^\circ$ $= 3058.31753869$ $PQ = \sqrt{3058.31753869}$ $= 55.30205727$ $= 55.3 \text{ km } 3 \text{ s } f$	C

MS-M6 :Non-right angled Trigonometry		Marks
Section II Solution		
Question 16		
(a)	$\tan 60^\circ = \frac{h}{640}$ $h = 640 \tan 60^\circ$ $= 1109 \text{ m}$	2
(b)	$\tan 65^\circ = \frac{d}{640}$ $d = 640 \tan 65^\circ$ $= 1372 \text{ m}$ Height Lighthouse = $1372 - 1109 = 264 \text{ m}$	2
Question 17		
(a)	(i) $\tan \theta = \frac{54}{50}$ $\theta = 47^\circ 12'$	1
(b)	(ii) $\cos 55^\circ = \frac{50}{x}$ $x = \frac{50}{\cos 55^\circ}$ $x = 87.2 \text{ m}$	2

Question 18

(a)	 $\frac{\sin \angle ADB}{16} = \frac{\sin 20}{14}$ $\sin \angle ADB = \frac{16 \sin 20}{14}$ $= 0.39088$ $\angle ADB = 23^\circ$	2
(b)	$\angle ADC = 70^\circ$ (angle sum $\triangle ACD$) $\therefore \angle BDC = 47^\circ$ ($70^\circ - 23^\circ$)	1
(c)	$\sin 47^\circ = \frac{BC}{14}$ $BC = 14 \sin 47^\circ$ $= 10.2 \text{ cm}$ $AC = 16 + 10.2$ $= 26.2 \text{ cm}$	2
Question 19		
(a)	$72 - 39 = 33 \quad 360 - 33 = 327^\circ$	1
(b)	$a^2 = b^2 + c^2 - 2bc \cos A$ $= 61^2 + 53^2 - 2 \times 61 \times 53 \times \cos 72 = 4531.896...$ $a = 67.319...m \cong 67m$	2

Question 20		
(a)	$YC^2 = 15^2 + 120^2$ $= 14625$ $YC = \sqrt{14625}$ $= 121 \text{ cm (3 s.f.)}$	2
(b)	$\text{Area} = \frac{1}{2}ab \sin C$ $= \frac{1}{2} \times 15 \times 121 \times \sin 95^\circ$ $= 904 \text{ cm}^2$	2
(c)	$\text{Area } \triangle ABC = \frac{1}{2} \times 120 \times 45 = 2700 \text{ cm}^2$ $\text{Percentage} = \frac{904}{2700} \times 100 = 33.5\%$	2
Question 21		
(a)	$\frac{YB}{\sin 34^\circ} = \frac{1.5}{\sin 86^\circ}$ $YB = \frac{1.5}{\sin 86^\circ} (\sin 34^\circ)$ $= 0.841 \text{ km}$	2
(b)	$XB^2 = 2.4^2 + 1.5^2 - 2 \times 2.4 \times 1.5 \times \cos 75^\circ$ $= 6.15$ $XB = 2.5 \text{ km}$	2
(c)	Bearing is $86^\circ + 14^\circ = 100^\circ \text{T}$ 	1

Question 22		
(a)	$\theta = 90 - 41 = 49^\circ$ $\beta = 90 - (20 + 49) = 21^\circ$	1
(b)	$\frac{BT}{\sin 20^\circ} = \frac{20}{\sin 21^\circ}$ $BT = \frac{20 \sin 20^\circ}{\sin 21^\circ}$ $= 19 \text{ m (nearest m)}$	2
(c)	$\sin 41^\circ = \frac{TC}{19}$ $TC = 19 \sin 41^\circ$ $= 12.5 \text{ m}$	2
Question 23		
(a)		1
(b)	$\angle KLM = 75 + 20 = 95^\circ$	1
(c)	$KM^2 = 120^2 + 140^2 - 2 \times 120 \times 140 \times \cos 95^\circ$ $= 36\,928$ $KM = 192 \text{ km}$	2
(d)	$\frac{\sin \angle LKM}{140} = \frac{\sin 95^\circ}{192}$ $\sin \angle LKM = \frac{140 \sin 95^\circ}{192}$ $\sin \angle LKM = 0.726$ $\angle LKM = 47^\circ$ $\text{Bearing} = 47^\circ + 75^\circ = 122^\circ$	2